Synopsis: AUTOSAR HLD Document Analysis Assistant

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Case study: Case Study 1, Automotive Engineering AI

Problem

AUTOSAR High-Level Design (HLD) documents are often hundreds of pages of unstructured PDF. Architects and testers find mismatches, missing dependencies and inconsistent terminology by reading and cross-checking by hand, which is slow and leaves no structured record.

Objective

Build a locally run assistant that extracts the architecture of an HLD into structured data, answers questions with page citations, checks the design for inconsistencies, and compares document versions, while keeping a human reviewer in control.

Intended users

System and AUTOSAR architects, software developers, integration engineers, test engineers and engineering managers. The prototype has three roles: viewer, engineer and admin.

Approach

A small language model reads the document once and fills a database of components, ports, interfaces and data elements. Questions with an exact answer (who provides a signal, a period, a unit, what changed) are answered from that database. Other questions use hybrid retrieval over the document text, a reranker, a refusal cutoff, a model answer

from numbered evidence, and a citation check. Rules find six kinds of inconsistency, and a graph database supports traceability queries. Every extracted row has a review status that engineers set in a web app.

Technology

Qwen2.5 3B served by Ollama, BGE embeddings, a cross-encoder reranker, ChromaDB, BM25, SQLite, Neo4j, FastAPI, Streamlit and Docker, with custom Python orchestration. Everything runs on a laptop CPU, with no cloud calls.

Data

A synthetic cruise-control HLD in two versions, with six planted design errors, because real project HLDs are confidential. Ground truth and test questions are stored with it.

Results so far

On the synthetic data, extraction scored precision and recall of 1.0, the checks found all six planted errors with no false positives, and the version comparison found all four changes. Cited question answering was fully correct on 10 of 14 new questions. All results come from one synthetic document, so they will probably overstate performance on real HLDs.

Limitations

No real HLD or public AUTOSAR specification was tested, diagrams inside PDFs are not interpreted, the question router is keyword based, and the small model is weak on absence and multi-step questions.